

Data Management Technologies

Unit 3 – Consolidation Task - Individual SQL Coding

/* DATABASE CREATION */

/* This deletes the table, so it is rewritten when the file is rerun */

DROP TABLE Books;

/*Create books table */

```
CREATE TABLE Books (  
    BookID INT PRIMARY KEY,  
    Title VARCHAR(100) NOT NULL,  
    Author VARCHAR(50),  
    Genre VARCHAR(30),  
    Price FLOAT  
);
```

/*Populate books data*/

```
INSERT INTO Books VALUES (1, 'To Kill a Mockingbird', 'Harper Lee', 'Fiction', 14.99);  
INSERT INTO Books VALUES (2, 'Tell Freedom', 'Peter Abrahams', 'Autobiography', 12.99);  
INSERT INTO Books VALUES (3, 'The Color Purple', 'Alice Walker', 'Historical Fiction', 9.49);  
INSERT INTO Books VALUES (4, 'Things Fall Apart', 'Chinua Achebe', 'Historical Fiction', 7.99);  
INSERT INTO Books VALUES (5, 'Outlander', 'Diana Gabaldon', 'Historical Fiction', 19.99);  
INSERT INTO Books VALUES (6, 'Kindred', 'Octavia Butler', 'Science Fiction', 10.99);  
INSERT INTO Books VALUES (7, 'A Tree Grows in Brooklyn', 'Betty Ross', 'Autobiography', 9.99);  
INSERT INTO Books VALUES (8, 'Animal Farm', 'George Orwell', 'Dystopian', 6.49);  
INSERT INTO Books VALUES (9, 'A Little Princess', 'Frances Hodgson Burnett', 'Fiction', 13.99);  
INSERT INTO Books VALUES (10, 'A Tale of Two Cities', 'Charles Dickens', 'Fiction', 7.99);
```

/*Print books data*/

```
SELECT * FROM Books
```

/* QUERY DATA */

/* 1) BASIC SELECT*/

/* Retrieve all data from Books table */

```
SELECT * from Books;
```

/*Retrieve only title and author columns */

```
SELECT title, author from Books;
```

/* 2) CONDITIONAL SELECT*/

/*Retrieve Books with Price above £15 */

```
SELECT * from Books WHERE price > 15;
```

/* Retrieve books in fiction genre */

```
SELECT * FROM Books WHERE (genre = 'Fiction') OR (genre = 'Historical Fiction') OR (genre = 'Science Fiction');
```

```
/* 3) SORTING RESULTS*/
```

```
/* Return all books in descending order */
```

```
SELECT * FROM Books ORDER BY price DESC;
```

```
/* MODIFY DATA */
```

```
/* 1) Update - Modifying specific information */
```

```
UPDATE Books SET price = 21.99 WHERE BookID = 1;
```

```
/* 2) Delete - Removing an entry */
```

```
DELETE FROM Books WHERE genre = 'Dystopian';
```

```
/* ADVANCED QUERIES */
```

```
/* 1) String Matching */
```

```
SELECT * FROM Books WHERE title LIKE '%kill%';
```

```
/* 2) Aggregation */
```

```
/* Average private price of all books */
```

```
SELECT AVG(price) FROM Books;
```

```
/* Total number of books in the Fiction genre*/
```

```
SELECT count(*) FROM Books WHERE genre = 'Fiction';
```

```
/*CREATING RELATIONSHIPS */
```

```
/* a) Adding Customers table*/
```

```
DROP TABLE BookCustomer;  
CREATE TABLE BookCustomer (  
    CustomerID INT PRIMARY KEY,  
    Name VARCHAR(50),  
    Email VARCHAR(50),  
    Phone VARCHAR(15)  
);
```

```
/* b) Adding Sales Table */
```

```
DROP TABLE BookSales;  
CREATE TABLE BookSales (  
    SaleID INT PRIMARY KEY,  
    BookID INT,  
    CustomerID INT,  
    Quantity INT,  
    FOREIGN KEY (BookID) REFERENCES Books(BookID),  
    FOREIGN KEY (CustomerID) REFERENCES BookCustomer(CustomerID)
```

);

/* c) Populate Data */

```
INSERT INTO BookCustomer VALUES (1, 'John Smith', 'johnny1989@gmail.com', '07845 672 384');
INSERT INTO BookCustomer VALUES (2, 'Gladys Ian', 'glady983@yahoo.com', '07734 549 234');
INSERT INTO BookCustomer VALUES (3, 'Robert Jones', 'RJ1234@outlook.com', '07732 345 294');
INSERT INTO BookCustomer VALUES (4, 'Mary Delacroix', 'MarDe@gmail.com', '08023 455 332');
INSERT INTO BookCustomer VALUES (5, 'James Johnson', 'james234@hotmail.com', '07743 263 472');
INSERT INTO BookCustomer VALUES (6, 'Sally Telford', 'sallytelly@yahoo.com', '07654 124 565');
INSERT INTO BookCustomer VALUES (7, 'Samuel Alton', 'samuel.alton@gmail.com', '07734 345 214');
INSERT INTO BookCustomer VALUES (8, 'Janet Crane', 'janecrane31@yahoo.com', '07756 326 952');
INSERT INTO BookCustomer VALUES (9, 'Luke Drew', 'lukasdukas@gmail.com', '07712 495 135');
INSERT INTO BookCustomer VALUES (10, 'Martha Hudson', 'martha-hudson@aol.com', '07834 412945');
```

```
INSERT INTO BookSales VALUES (001, 1, 3, 1);
INSERT INTO BookSales VALUES (002, 1, 4, 2);
INSERT INTO BookSales VALUES (003, 4, 5, 1);
INSERT INTO BookSales VALUES (004, 8, 2, 1);
INSERT INTO BookSales VALUES (005, 3, 4, 1);
INSERT INTO BookSales VALUES (006, 8, 4, 2);
INSERT INTO BookSales VALUES (007, 9, 1, 1);
INSERT INTO BookSales VALUES (008, 2, 2, 1);
INSERT INTO BookSales VALUES (009, 5, 3, 2);
INSERT INTO BookSales VALUES (010, 7, 4, 1);
INSERT INTO BookSales VALUES (011, 6, 8, 1);
INSERT INTO BookSales VALUES (012, 9, 9, 1);
```

/* Complex Queries */

/* List all books sold to a specific customer */

```
SELECT * FROM BookSales WHERE CustomerID = 4;
```

/* Calculate total sales revenue for each book */

```
SELECT Books.title, SUM(Books.price * BookSales.Quantity) AS TotalRevenue
FROM BookSales
JOIN Books ON BookSales.BookID = Books.BookID
GROUP BY Books.BookID;
```

Comments on issues:

- INSERT INTO commands can only be run once without returning an error message. Adding DROP table before CREATE table can help to avoid this by deleting and recreating the table each time the code is run.